

PRIYANSHU MITRA

Software Engineer | AI Automation Engineer, | Robotics & Automation Engineer

Delhi, India

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PROFESSIONAL SUMMARY

Software Engineer and Robotics Automation Engineer with experience in software development, AI automation, robotics, IoT systems, and data analytics. Skilled in Python, C++, JavaScript, React, Node.js, embedded systems, and machine learning fundamentals. Experienced in developing intelligent software solutions, robotics applications, and scalable systems while mentoring students and leading technical projects.

EDUCATION

Institute of Technology and Science, India	<i>October 2022 – June 2025</i>
Bachelor of Computer Applications (BCA), Computer Science & Technology	
Mukherjee Memorial Senior Secondary School, India	<i>April 2021 – March 2022</i>
12th Standard (CBSE Board)	
Mukherjee Memorial Senior Secondary School, India	<i>April 2019 – March 2020</i>
10th Standard (CBSE Board)	

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, Kotlin

Software Engineering: Data Structures & Algorithms, OOP, SDLC, REST APIs, System Design, Debugging, Software Testing

Frontend Development: HTML, CSS, JavaScript, React.js

Backend Development: Node.js, Express.js, Typescript

Databases: MySQL, MongoDB

Robotics & IoT: Arduino, ESP32, Raspberry Pi, Embedded Programming, Robotics Automation, ROS2, Gazebo, Sensors, Actuators and multi-types of Microcontroller

AI & Data Analytics: AI Automation, Machine Learning & AI Fundamentals, Dataset Filtration with Handling, Microsoft Office, Power BI, Google Workspace, Prompt Engineering, Dashboard Building

Tools: Git, GitHub, Android Studio, N8N, Shopify

Soft Skills: Communication, Problem Solving, Decision Making, Team Collaboration, Technical Mentoring, Project Management

PROFESSIONAL EXPERIENCE

Robotics Engineer & Trainer (Full Time)	<i>February 2026 – Present</i>
<i>KalkiOrion Robotics Labs, New Delhi, India</i>	

- Developed 80+ robotics and embedded systems projects, including Advanced robotic arms, autonomous bots, Multi-Types of Drones, IoT Devices and many more using sensors, Actuators and microcontrollers in this firm.
- Conducted hands-on training sessions on robotics, programming, embedded systems, and AI concepts for 500+ students across schools and Institutes, enhancing their practical and analytical skills.
- Mentored students in robotics competitions and innovation challenges, guiding complete project development from ideation to functional prototypes with a high project success rate.
- Designed structured robotics curriculum, technical workshops, and live demonstrations focused on real world engineering applications, automation, and industry-oriented project development.

Robotics Engineer & Trainer (Full Time)	<i>August 2025 – January 2026</i>
<i>Techradiance, New Delhi, India</i>	

- Designed and developed 100+ robotics projects including robotic arms, obstacle-avoiding bots, line followers, and AI-based systems using sensors, microcontrollers, and automation tools.
- Delivered technical training on robotics, programming, and AI fundamentals to 900+ students across schools, improving their problem-solving and logic-building capabilities.
- Led robotics teams to 95

Software Developer (Full Time)	<i>January 2025 – August 2025</i>
<i>Oasis Infobyte, New Delhi, India</i>	

- Developed and deployed 10+ software applications and web solutions using Python, C++, JavaScript, React, Node.js, and database technologies.
- Built RESTful APIs, automated workflows, and data-driven applications, improving operational efficiency and reducing manual effort.
- Designed responsive user interfaces and scalable backend systems while ensuring code quality through testing, debugging, and optimization.
- Collaborated with clients and development teams to gather requirements, implement features, and deliver projects within deadlines.

Data Analyst & Technical Specialist (Intern)

November 2024 – January 2025

Peppy Cartel Pvt. Ltd., New Delhi, India

- Extracted, cleaned, and analyzed large datasets using Python, SQL, and Excel to generate actionable insights that improved business decision-making and performance metrics by 25%.
- Collaborated with cross-functional teams to identify technical requirements and delivered efficient data driven solutions within tight deadlines.
- Received Letter of Recommendation from HR for outstanding performance, dedication, and technical problem-solving abilities.

PROJECTS

Autonomous AI Robotics & Navigation System :

- Designed and developed an autonomous mobile robot integrating ROS2, C++, Python, computer vision, ultrasonic/IR sensors, and real-time motor control for intelligent navigation.
- Implemented obstacle detection, path-following, sensor processing, and autonomous decision-making algorithms for real-time robotic movement.
- Simulated and tested robotic behaviors using robotics simulation tools, optimizing navigation and control algorithms for reliable operation.

Vision-Guided Robotic Arm & Manipulation System :

- Developed a vision-guided robotic arm integrating computer vision, Python/C++, ROS2, sensors, and servo-based actuation for automated object detection and manipulation.
- Implemented object detection, coordinate estimation, inverse kinematics, and motion-control logic to enable autonomous pick-and-place operations.
- Integrated real-time perception and robotic control pipelines to improve precision, responsiveness, and autonomous task execution.

Multilingual AI Knowledge Retrieval & Semantic Search Platform :

- Developed an AI-powered knowledge retrieval platform using Python and NLP to search and retrieve information across multiple languages.
- Implemented semantic search, document indexing, contextual ranking, and intelligent query matching for efficient information discovery.
- Designed scalable backend services and data-processing pipelines for handling large knowledge repositories and multilingual queries.
- Integrated AI-driven retrieval and contextual response generation to improve search relevance and user experience.

Real-Time Robotics & IoT Monitoring Platform :

- Developed a real-time monitoring platform for robotics and IoT systems to visualize sensor data, device status, system performance, and operational events.
- Implemented REST APIs, WebSocket-based real-time communication, database integration, authentication, and responsive dashboards using modern web technologies.
- Integrated microcontrollers and connected devices with backend services for real-time telemetry, event monitoring, and automated alerts.

CERTIFICATIONS

Microsoft: SDE, Data Analytics

Deloitte: Deloitte Australia-Job Simulation in Software Engineering

Nvidia: Robotics Fundamentals with Jetson NANO